Feature Stores, Data Pipelines & Offline/Online Serving

Scaler School of Technology | ML System Design and LLMOps | Week 2 — Session 1/2 | Pre-Lecture Prep

What Is This Lecture About?

Last session you saw the big picture: a production ML system has six components and many failure modes. Today we go deep on the **foundation**: data. Every model is only as good as the data it trains on and the features it serves with. This session covers how data is stored, moved, transformed into features, and made available consistently at both training and serving time.

Data as the Foundation

Where ML data comes from (user input, logs, internal databases, third-party), and how the format you choose — JSON, CSV, or Parquet — affects the speed and cost of every downstream step in your pipeline.

Pipelines That Move Data

The three ways services share data (databases, REST/gRPC, real-time transport) and when to use batch vs. streaming processing. Kafka as a data broker: what it decouples and why that matters for real-time ML.

Feature Engineering Traps

The most dangerous mistakes in ML are invisible: data leakage from scaling before splitting, and using future information in training rows. We will see exactly why these inflate offline metrics while producing models that fail in production.

Feature Stores: Offline and Online

Why the same feature must be computed differently for training (historical, batch) vs. serving (real-time, low-latency), and how feature stores solve this by making you compute once and use everywhere.

What to Expect in the Session

- Opening discussion: where do your features come from right now?
- Data format comparison — hands-on feel for row-major vs. column-major
- Data leakage demo: the surprising accuracy you can get by doing it wrong
- Feature store architecture diagram and walkthrough
- Two short quizzes on formats, leakage, and batch vs. streaming
- Feast demo: defining, materializing, and retrieving a feature view

How to Prepare

- Recall the last ML project you worked on — where did your features come from? Were they computed the same way at training and serving time?
- Think about a feature like "user's average session length" — how would you compute this during training on historical data vs. during serving on a live request?
- Optional: read Chapter 3 of Chip Huyen's *Designing Machine Learning Systems* (Data Engineering Fundamentals) — it is the primary source for today's content
- Optional: skim the Feast documentation quickstart at feast.dev — we will use it in the demo
- Ensure you have Google Colab access; we will run a short notebook in session

Duration: ~2 hours **Prerequisites:** Week 1 Session 1 (ML Systems 101) **Instructor:** Priyansh Saxena